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Temperature dependence of the deformation behavior and mechanical properties of a Fe–Mn–Al–C low-density steel for cryogenic application

Qi Zhang [a,b], Guanghui Chen [a,b,*], Haijiang Hu [a,b,**], Wanjie Zheng [a,b], Xiangliang Wan [a,b], Zhengliang Xue [a,b], Guang Xu [a,b]

[a] State Key Laboratory of Refractories and Metallurgy, Wuhan University of Science and Technology, Wuhan, Hubei, 430081, PR China
[b] Key Laboratory for Ferrous Metallurgy and Resources Utilization of Ministry of Education, Wuhan University of Science and Technology, Wuhan, Hubei, 430081, PR China

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ABSTRACT

The temperature dependence of the deformation behavior and mechanical properties of a Fe–20Mn–6Al-0.6C-0.15Si austenitic low-density steel was studied by tensile test and Charpy impact test at −196 °C, −77 °C and room temperature (RT), followed by microstructure analyses. The results indicate that the decreased tensile temperature, which is corresponding to reduced stacking fault energy (SFE), promotes the planar glide of dislocations. This led to larger fraction of low-angle grain boundaries (LAGBs) at lower tensile temperature. Consequently, both the tensile strength and elongation were improved as temperature decreased. At lower impact test temperature, the load and energy of crack initiation were larger, indicating a higher resistance to crack initiation. However, the transition from stable to unstable crack propagation occurred more rapidly at lower temperature, resulting in lower crack propagation energy and reduced resistance to crack propagation. This is because less region experienced planar glide. Due to the rapid unstable crack propagation, the Charpy absorbed energy at −196 °C was the lowest. The smaller fracture surface area, larger fibrous zone and more obvious ductile fracture characteristics in radial zone suggest that the toughness is better at higher temperature. Additionally, the presence of secondary cracks at RT delayed fracture, therefore benefitting toughness.

1. Introduction

Fe–Mn–Al–C austenitic low-density steels exhibit light weight and outstanding mechanical properties. The density could be reduced to 1.3% with the addition of 1.0 wt% Al [1,2]. At room temperature (RT), the tensile strength, total elongation and impact toughness could reach 1500 MPa, 80% and 200 J/cm^2, respectively [2–4]. Therefore, Fe–Mn–Al–C austenitic low-density steels are widely regarded as ideal materials for the automotive industry. Other austenitic steels, such as austenitic stainless steels and twinning-induced plasticity (TWIP) steels, typically display great combination of strength and ductility during cryogenic deformation [5–8]. Consequently, Fe–Mn–Al–C austenitic low-density steels are expected to have broad application prospects in cryogenic applications, such as cryogenic components of spacecraft and ships, transportation and storage of liquefied natural gas (LNG) and liquefied petroleum gas (LPG) [9–12].

The deformation mechanism and mechanical properties of austenitic steels mainly depend on their stacking fault energy (SFE). For low-SFE materials such as austenitic stainless steels and Fe–Mn–C TWIP steels, deformation-induced martensitic transformation (DIMT) and deformation twinning are the dominant factors contributing to the high work hardening rate, high strength and good ductility [13–16]. In high-SFE materials, perfect dislocations are easy to glide from one slip plane to another, a process known as cross slip [17]. For Fe–Mn–Al–C austenitic low-density steels, their SFE usually reaches a relatively high level due to high Al content [18]. Such a high SFE could effectively suppress DIMT and deformation twinning. Many literatures [19–21] reported that the plastic deformation of Fe–Mn–Al–C austenitic low-density steels is still accompanied with the features of planar glide instead of wavy glide in spite of a high SFE. For instance, Yoo and Park [20] proposed that micro-bands and their intersections mainly occurred in Fe–28Mn–9Al-0.8C (concentration in wt.% unless otherwise specified) austenitic low-density steel with an SFE of 85 mJ/m^2 during tensile deformation at RT, i.e., the deformation mechanism was dominated by

* Corresponding author. State Key Laboratory of Refractories and Metallurgy, Wuhan University of Science and Technology, Wuhan, Hubei 430081, PR China.
** Corresponding author. State Key Laboratory of Refractories and Metallurgy, Wuhan University of Science and Technology, Wuhan, Hubei 430081, PR China.
E-mail addresses: chenguanghui@wust.edu.cn (G. Chen), huhaijiang@wust.edu.cn (H. Hu).

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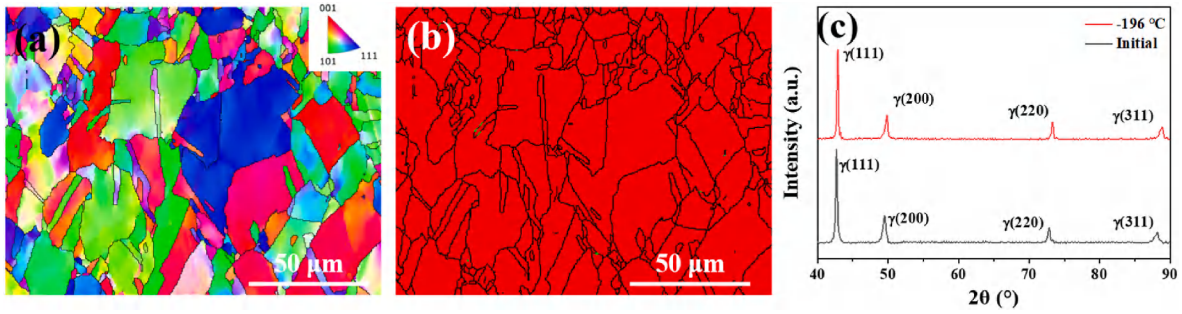


Fig. 1. EBSD IPF map (a) and phase map (b) of initial steel; (c) XRD pattern of the initial steel and after holding at −196 °C for 30 min.

micro-band induced plasticity (MBIP). Welsch et al. [22] observed a similar deformed microstructure in Fe–30Mn–8Al-1.2C austenitic steel with SFE of 85 mJ/m^2, which was also characterized by slip bands and slip bands were dynamically refined with strain, a mechanism known as dynamic slip band refinement (DSBR). Similar planar glide features after deformation at RT were also reported in other Fe–Mn–Al–C austenitic steels with higher SFE [23,24]. Many studies about austenitic Fe–Mn–Al–C low-density steels have mainly focused on their deformation mechanism and mechanical properties at RT, which are highly relevant to applications in the automotive industry. Considering the relatively high level of SFE at RT, reducing the SFE of austenitic low-density steels is expected to lead to better mechanical properties at cryogenic temperatures. Chen et al. [10] suggested that Fe-0.6C–18Mn-0.5Si-(3–5)Al austenitic low density steels exhibited good cryogenic impact toughness due to the formation of mechanical twins and dislocation walls, while 8 wt% Al addition leading to ferrite formation reduced the cryogenic impact toughness. However, Luo and Gan [25] reported that the impact toughness of as-aged Fe–28Mn–5Al–1C austenitic alloy reduced with the decreasing temperature from 27 to −196 °C because fracture surface becomes mainly intergranular fracture at temperature below −90 °C due to κ-carbides.

In this paper, with the aim of investigating the temperature dependence of deformed microstructures, work hardening behavior and mechanical properties, tensile tests and Charpy impact tests were conducted at different temperatures for an Fe–20Mn–6Al-0.6C-0.15Si austenitic low-density steel. The results provide theoretical support for the cryogenic application of austenitic low-density steels.

2. Experimental material and methods

The chemical composition of the austenitic low-density steel is Fe–20Mn–6Al-0.6C-0.15Si. The experimental steel was melted and cast in a 50 kg vacuum induction furnace and then air cooled to RT. The ingot was then reheated to 1250 °C and held for 2 h for homogenization. Subsequently, the ingot was forged into round bars with a diameter of 60 mm followed by cooling to RT in the furnace. The starting and finishing forging temperatures were 1100 and 950 °C, respectively.

The initial microstructure of the forged steel was analyzed using electron backscatter diffraction (EBSD) on an Apreo S HiVac scanning electron microscope (SEM) and a SmartLab SE X-ray diffractometer (XRD) with Cu Kα radiation (λ = 0.154060 nm), respectively. Another sample was held at liquid nitrogen for 30 min followed by XRD analysis.

Tensile tests were carried out on a YYF-50 tensile tester until fracture. Liquid nitrogen (−196 °C) and a dry ice-alcohol mixture (−77 °C) were used as cooling medium. The samples were tensile deformed at −196 °C, −77 °C and RT, respectively. The entire tensile process was

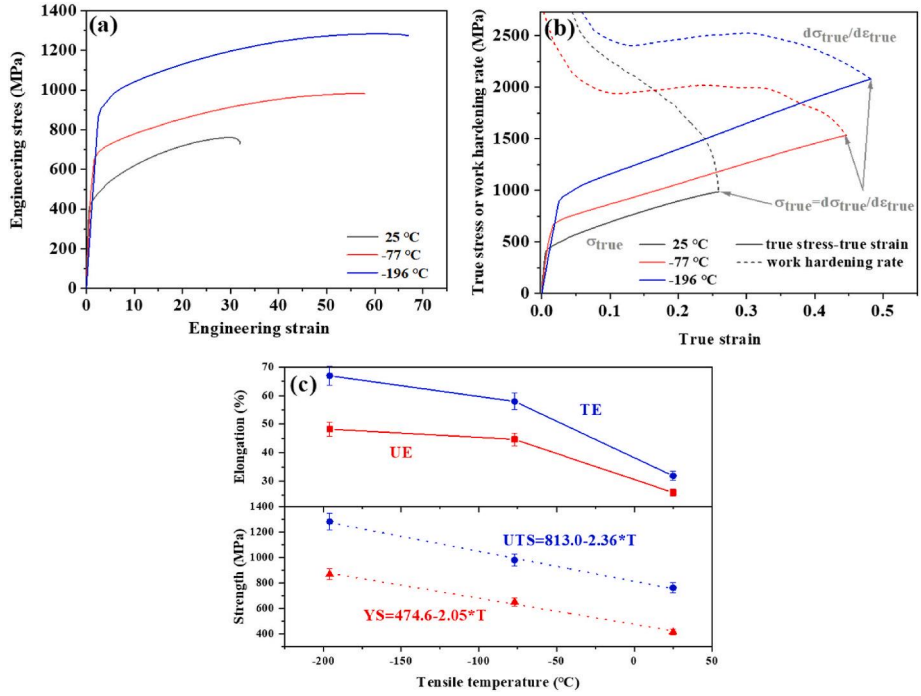


Fig. 2. Tensile test results at −196 °C, −77 °C and RT: (a) Engineering stress-strain curves; (b) true stress-strain curves and work hardening rate curves; (c) elongation and strength varying with temperatures.

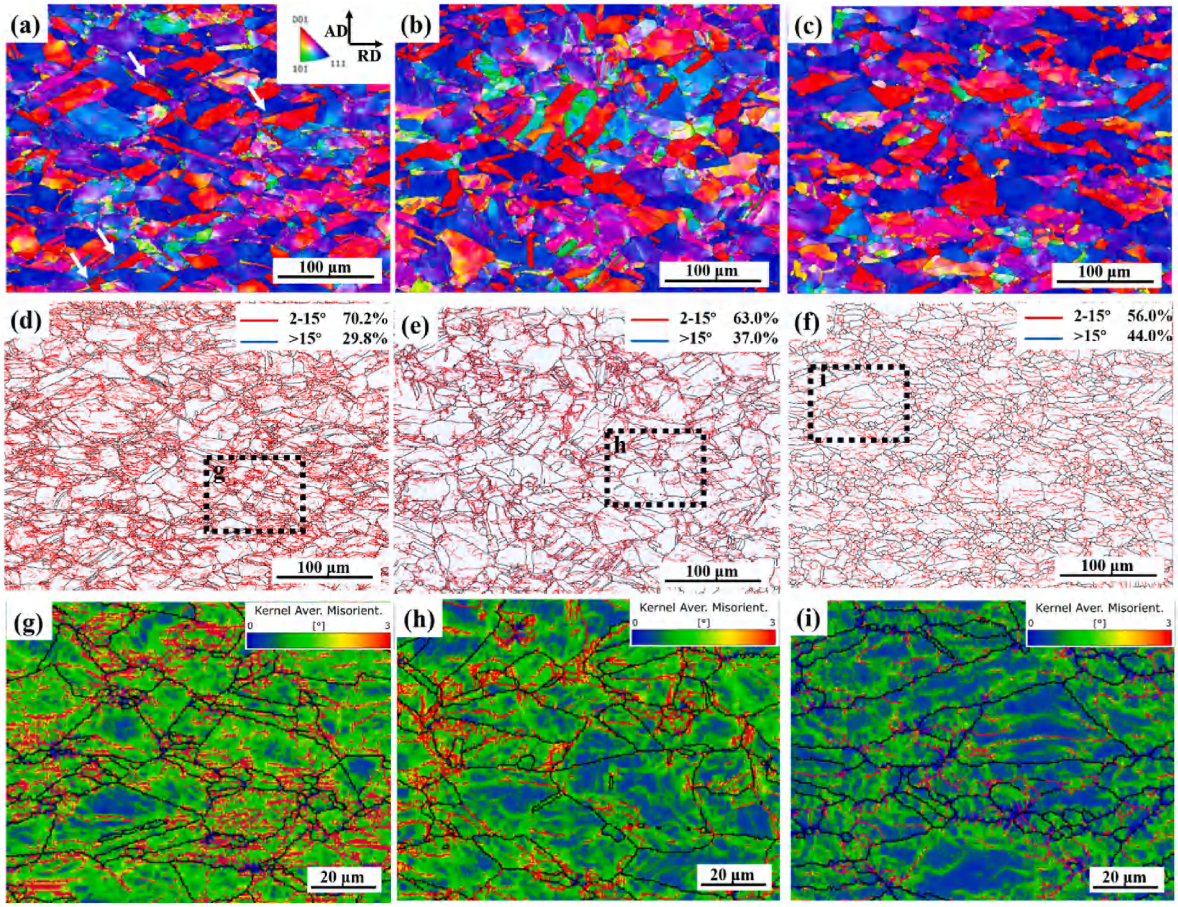


Fig. 3. EBSD analyses for the experimental steel deformed at different temperature: (a,d,g) −196 °C; (b,e,h) −77 °C; (c,f,i) RT. (a–c) IPF maps; (d–f) GB maps; (g–i) GND distribution maps superposed with GB maps for the region marked with frame in (d–f). In IPF maps, the colors indicate crystal directions parallel to the tensile direction. Mechanical twins are marked with arrows. In grain boundary distribution maps, the fractions of LAGBs and HAGBs are marked with red and black colors.

carried out in cooling medium. Additionally, the adopted strain rate of 0.001 s–1 was slow enough to avoid frictional heat. Three tensile samples with a parallel length of 32 mm, width of 5 mm and thickness of 1.5 mm were tested for each condition to get the average values. The adopted strain rate was 10^{-3} s^{-1}. An extensometer was used during tensile tests. The fracture surface after tensile tests was observed by a Nova 400 Nano SEM. The uniformly deformed sections, which refers to the deformed region not affected by necking, approximately 10 mm from the fracture were mechanically polished to eliminate the deformation layer formed during specimen preparation. Then the deformed microstructure was analyzed by EBSD on an Apreo S HiVac SEM. The step size for EBSD analyses was 0.5 μm. Additionally, the deformed microstructure was observed by a TecnailF30 field emission transmission electron microscopy (TEM). The specimens for TEM observation were subjected to double-jet thinning by a Struers TenuPol-5 Twin-Jet Electropolisher. The electrolyte composition is 6% perchloric acid and 94% glacial acetic acid.

Charpy impact experiments were conducted at −196 °C, −77 °C and RT, respectively, on a PST752H-4 WANCE impact tester. The samples with the size of 10 × 10 × 55 mm with a 2 mm depth V-notch were used for Charpy impact experiments. The load-deflection curves were obtained from the Charpy impact tester. The fracture surface after the Charpy impact tests was observed by VHX-5000 ultra-depth of field three-dimensional (3D) microscope and SEM. In order to analyze the microstructure of the cross-section near the fracture surface, argon ion-beam cross-section polishing was performed on a Gatan/685.C automated argon ion polishing system. Then the microstructure was analyzed by EBSD. The step size for EBSD analyses was 0.8 μm.

3. Results

3.1. Initial microstructure

The density of the experimental steel (ρ_γ, g/cm^3) can be estimated according to equation (1) proposed by Kim et al. [19].

$$\rho_\gamma = 8.10 - 0.101(Al\%) - 0.41(C\%) - 0.0085(Mn\%) \quad (1)$$

The estimated density is 7.078 g/cm^3 and the measured density is 7.114 g/cm^3. Fig. 1a and b presents the EBSD inverse pole figure (IPF) map and phase map of the initial as-forged steel. The initial microstructure consists of equiaxed grains with fully austenitic constituent. The XRD patterns of as-forged steel and as-treated at −196 °C shown in Fig. 1c also suggest that the initial microstructure is fully austenitic and no isothermal martensite forms at −196 °C, indicating a very stable austenite.

3.2. Tensile tests

The experimental steel was subjected to tensile deformation at −196 °C, −77 °C and RT until fracture, respectively. The engineering stress-strain curves are plotted in Fig. 2a. Fig. 2b shows the corresponding true stress-strain curves and work hardening rate curves. It is observed that the work hardening rate decreases dramatically with strain at the initial stage of deformation, corresponding to the transition from elastic deformation to plastic deformation [5,16,26]. Subsequently, the experimental steels deformed at −77 and −196 °C show a near-linear work hardening rate with strain, but a continuous decrease is

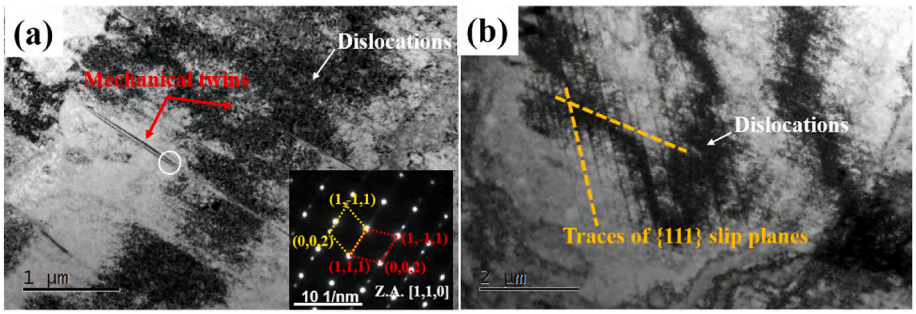


Fig. 4. TEM analysis results: (a) TEM image of the sample tensile deformed at −196 °C and corresponding SAED pattern; (b) microbands of the sample tensile deformed at −77 °C.

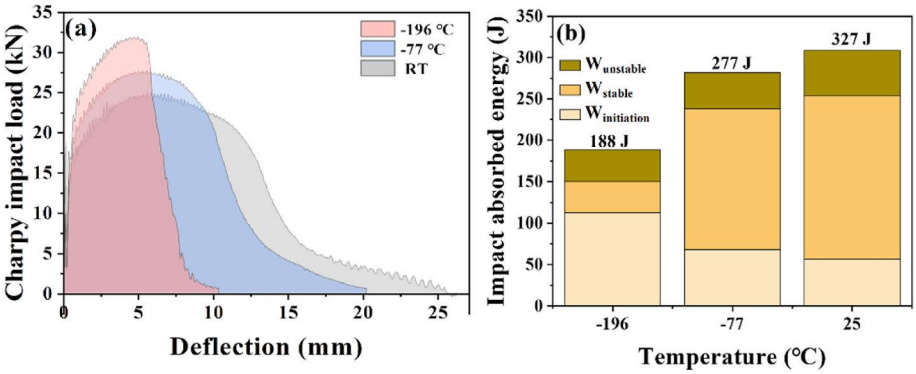


Fig. 5. Charpy impact load (a) and Charpy impact absorbed energy (b) of experimental steel.

observed at RT. The decreasing work hardening at RT has also been reported for other austenitic low-density steels [2]. In addition, as the deformation temperature decreases, the work hardening rate increases, indicating a negative temperature dependence of strength. The true strain at which true stress (σ_{true}) equals to work hardening rate ($d\sigma_{true}/d\varepsilon_{true}$) corresponds to the necking point. The tensile properties of the experimental steel at different temperatures are summarized in Fig. 2c. It is evident that the total elongation (TE) and uniform elongation (UE) increase with decreasing temperature. Additionally, decreasing deformation temperature leads to linear increase in both yield strength (YS) and ultimate tensile strength (UTS).

3.3. Tensile deformed microstructure

The tensile deformed microstructure of the uniform deformation section was analyzed by EBSD. Fig. 3 presents the IPF maps and grain

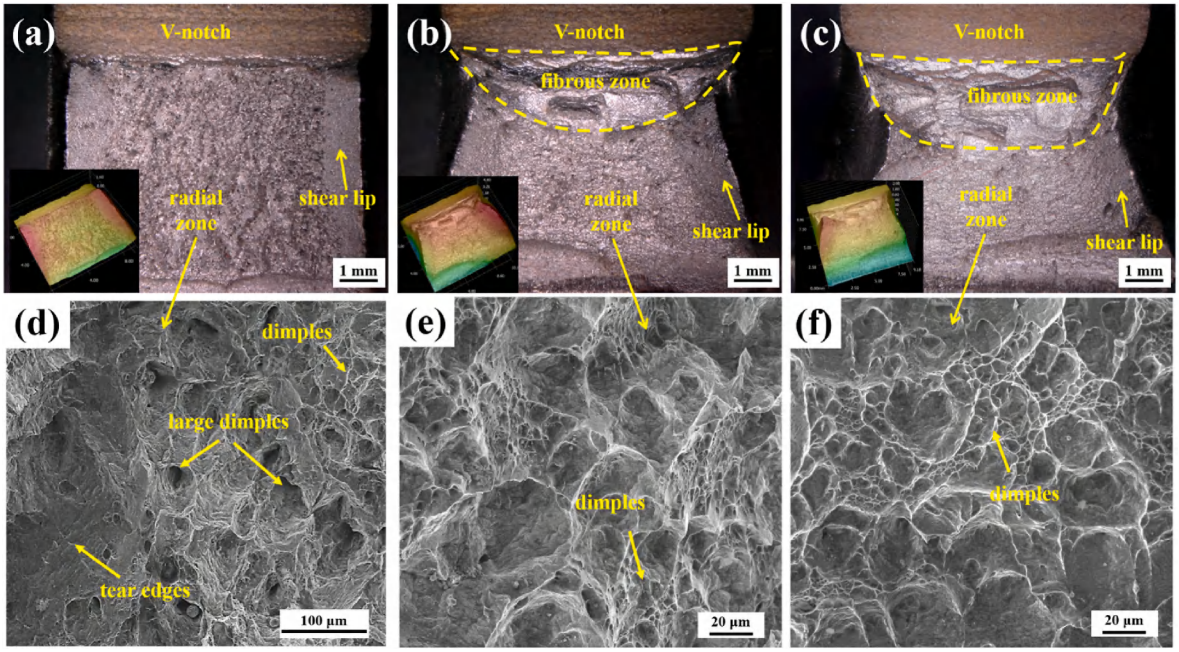


Fig. 6. Fracture surfaces of the impact samples tested at different temperatures: (a,d) −196 °C; (b,e) −77 °C; (c,f) RT.

boundary (GB) distribution maps for the tensile deformed microstructure. The IPF maps as presented in Fig. 3a-c shows that the crystal orientation of deformed austenite grains concentrates in {111} and {001}, which is a result of coordinated plastic deformation. However, as deformation temperature decreases to −196 °C, the decreasing SFE leads to small amount of mechanical twins formed during tensile deformation, as marked with arrows in Fig. 3a. The occurrence of mechanical twins is confirmed by TEM analysis as shown in Fig. 4a. It is seen that the mechanical twinning plane formed along <112> crystal direction on {111} plane in the case of the grain oriented near <110> zone axis. Compared with the sample deformed at −196 °C, no mechanical twins but microbands are observed in the sample deformed at −77 °C, as shown in Fig. 4b. A typical characteristic of planar glide that the uniform and dense distribution of microbands present in the deformed microstructure, which has been widely reported in austenitic low density steels [27,28]. The corresponding selected area electron diffraction (SAED) pattern indicates that these bands are orientated along {111} planes. The fractions of high-angle grain boundaries (HAGBs) and low-angle grain boundaries (LAGBs) are marked in the grain boundary distribution maps in Fig. 3d-f. It can be seen that the LAGBs mainly concentrate at these austenite grain with finer grain size. Additionally, the fraction of LAGBs dramatically increases with the decreasing temperature. The geometrically necessary dislocation (GND) distribution maps superposed with GB maps for the region marked with frame in Fig. 3d-f are shown separately in Fig. 3g-i. It is seen that the GNDs accumulate at LAGBs, indicating a more severe local deformation of these regions.

3.4. Impact tests

Fig. 5a shows the load-deflection curves obtained by instrumented Charpy impact test at different temperatures. Fig. 5b illustrates the changes of Charpy impact absorbed energy (W_{all}) with the test temperatures. The W_{all} at RT is about 327 J. As the test temperatures decreases to −77 and −196 °C, the W_{all} reduces to 277 and 188 J, respectively. Lower W_{all} values at lower temperature have also been reported by Wang et al. [11]. The activation energy for slip systems increases with the decreasing temperatures [29]. It becomes more difficult to activate planar slip systems at lower temperatures, leading to early fracture. Therefore, the W_{all} becomes lower at cryogenic temperature.

Fig. 6 provides fracture surface of the experimental steel tested at different temperatures. Shear lips are present on the fracture surface of all samples (Fig. 6a-c), which is independent of test temperature, indicating ductile fracture. Additionally, it is obvious that all the samples were compressed, leading to a reduction in the fracture surface area, which becomes more pronounced as temperature increases, indicating that the sample underwent more severe plastic deformation at a higher temperature. Almost no fibrous zone can be observed on the sample tested at −196 °C, while a larger fibrous zone appears at −77 °C and RT, suggesting that the experimental steel exhibits a better toughness at −77 °C and RT. The inserted 3D topography maps in Fig. 6a-c suggest that the fracture surface of the impact sample at −196 °C appeared significantly flatter compared to others. The morphology of radial zone was additionally observed using SEM (Fig. 6d-f). Fig. 6a-c obviously suggests that the fracture surface of the impact sample at −196 °C appeared significantly flatter compared to others. At −196 °C, the tested sample displayed a combination of ductile and brittle fracture. The fracture surfaces contain high-density large dimples and tear edges, as marked in Fig. 6d. These features indicate a tendency to be quasi-cleavage cracks [30], which is consistent with the relatively low W_{all} of 188 J. However, as the temperature increases, the morphology characteristics of ductile fracture become more obvious. Generally, the fracture morphology characteristics for the sample tested at −77 °C and RT demonstrate a more obvious ductile fracture, which is in good agreement with the result that the W_{all} is higher at −77 °C and RT.

Table 1
Thermodynamic parameter expressions and values (T is temperature, K) [6].

Thermodynamic parameters	Expressions/Values
$\Delta G_{Fe}^{\gamma\to\varepsilon}$	$-1828.4 + 4.686T$
$\Delta G_{Mn}^{\gamma\to\varepsilon}$	$3970–1.667T$
$\Delta G_{Al}^{\gamma\to\varepsilon}$	$5481.04–1.7991T$
$\Delta G_{C}^{\gamma\to\varepsilon}$	-22.166
$\Delta G_{FeMn}^{\gamma\to\varepsilon}$	$-9135.5 + 15.2821X_{Mn}$
$\Delta G_{FeAl}^{\gamma\to\varepsilon}$	3326.280
$\Delta G_{FeC}^{\gamma\to\varepsilon}$	42.500
$\Delta G_{FeMnAl/C}^{\gamma\to\varepsilon}$	$1246(1\text{-}e^{-24.29X_C})/X_c\text{-}17175X_{Mn}$

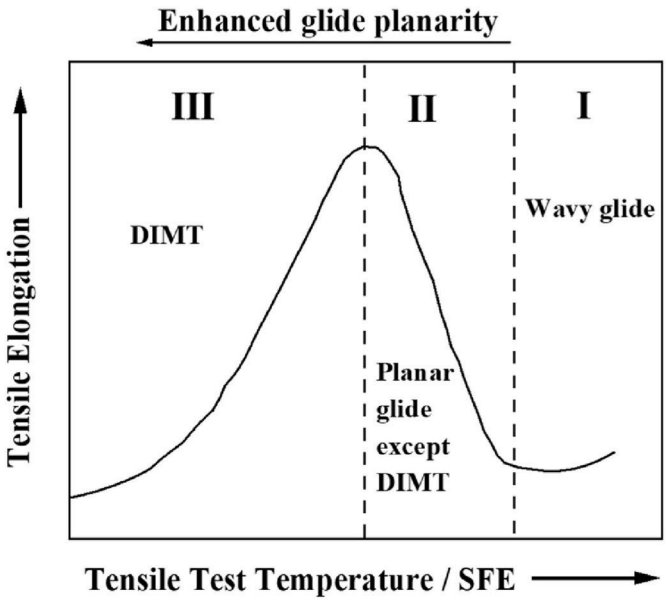


Fig. 7. Illustration for the temperature dependence of tensile elongation of austenitic steels [32].

4. Discussion

4.1. Deformation mechanisms

The SFE of the experimental steel can be calculated according to equation (2) [31]:

$$SFE = 2\rho_A \Delta G^{\gamma\to\varepsilon} + 2\sigma^{\gamma/\varepsilon} \tag{2}$$

where ρ_A refers to the molar atomic density on the $\{111\}_\gamma$ plane. The term $\Delta G^{\gamma\to\varepsilon}$ represents the free energy of transformation from austenite to ε-martensite, and $\sigma^{\gamma/\varepsilon}$ is the surface energy of the interface, which is taken as 15 mJ/m^2. The term ρ_A is expressed using equations (3) and (4):

$$\rho_A = \left(\frac{4}{\sqrt{3}}\right) \bullet \left(\frac{1}{N\alpha_\gamma{}^2}\right) \tag{3}$$

where N is Avogadro constant, and α_γ is the lattice parameter of austenite, calculated to be 0.36757 nm based on XRD analysis. The term $\Delta G^{\gamma\to\varepsilon}$ in equation (2) can be expressed by equation (4) [6]:

$$\Delta G^{\gamma\to\varepsilon} = \sum X_i \Delta G_i^{\gamma\to\varepsilon} + \sum X_i X_j \Delta G_{ij}^{\gamma\to\varepsilon} + X_c \Delta G_{FeMnAl/C}^{\gamma\to\varepsilon} + \Delta G_{mg}^{\gamma\to\varepsilon} \tag{4}$$

where X_i and X_j represent the mole fractions of different alloying elements, $\Delta G_i^{\gamma\to\varepsilon}$ represents the difference in free energy between the face-centred cubic structure and the densely arranged hexagonal structure of a given alloying element, $\Delta G_{ij}^{\gamma\to\varepsilon}$ represents the difference in the interaction energy of two alloying elements, and $\Delta G_{mg}^{\gamma\to\varepsilon}$ represents the

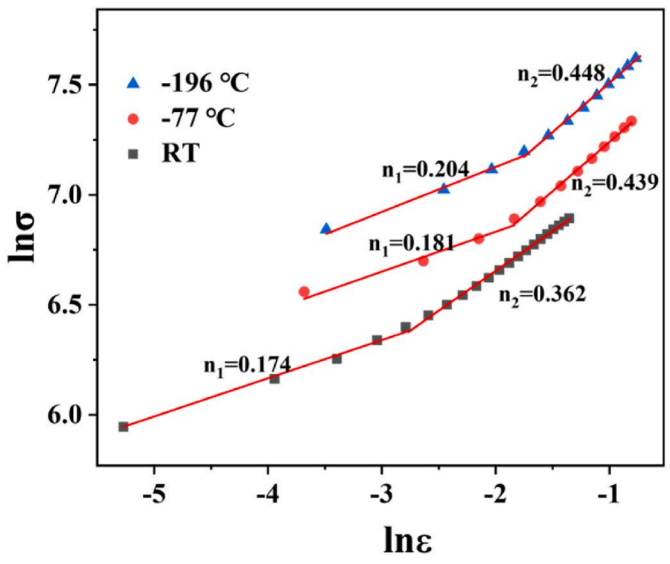


Fig. 8. Determination of work hardening exponent at different temperature.

magnetic free energy. The thermodynamic expressions and required parameters for equation (4) are listed in Table 1. The SFE of the experimental steel at −196 °C, −77 °C and RT is estimated to be 32, 40 and 61 mJ/m^2, respectively.

It is widely accepted that the mechanical properties of materials depend on the plastic deformation mechanism of austenite, which is strongly SFE-dependent [5,16]. Fig. 7 illustrates the generalized temperature (or SFE) dependence of the tensile elongation of austenitic steels [32]. At high SFE (region I), the three-dimensional mobility of perfect dislocations is enhanced. The dominant deformation mechanism is wavy glide, which leads to the formation of dislocation cells [33]. The dislocation cells cause strain localization and stress concentration, thus resulting in a low elongation. When SFE is lower (region II), perfect dislocations separate into two extended Shockley dislocations with wide separation [34]. In such a case, dislocations have a low three-dimensional mobility, which means that the dislocations slipping is restricted on a single slip plane [17,35]. In region II, the SFE is so low that wavy glide is absent but not low enough to initiate DIMT. In the relatively higher SFE range in region II, mechanical twins are usually formed as shown in Fig. 4a. At relatively lower SFE range, the dislocation planar slip results in the formation of glide bands as shown in Fig. 4b. The product of planar glide such as mechanical twins and glide bands could act as barriers for the movement of partial dislocations (Fig. 4), thus contributing to a high work hardening rate. Dislocation planar glide is easier than wavy glide at low temperature. The lower the deformation temperature, the higher the glide planarity [16]. High glide planarity is conducive to the dislocation multiplication and movement, and therefore continuous uniform deformation [36]. Therefore, the tensile elongation increases with the decreasing SFE. Planar glide except DIMT dominates the deformation mechanism of these steels with moderate SFE such as TWIP steels [37–39] and austenitic low-density steels [27,40,41]. When the SFE is in region III, DIMT is the dominant deformation mechanism. The occurrence of DIMT leads to a high strength but also a brittleness. Lower SFE promotes DIMT, therefore the elongation decreases with the decreasing temperature.

Normally, for high Mn steels, when the SFE is between 18 and 35 mJ/m^2 [42], mechanical twinning is the main deformation mechanism. Outside this range, DIMT and dislocation planar glide dominate deformation mechanism [42]. The SFE of the experimental steel at RT and −77 °C is high enough that DIMT and mechanical twinning can be validly suppressed. When the deformation temperature decreases to −196 °C, a SFE of 32 mJ/m^2 could still totally suppress DIMT but enable the occurrence of mechanical twinning. Therefore, as deformation temperature decreases from RT to −196 °C, the decreasing SFE promotes dislocation planar glide, resulting in the increasing strength and elongation (Fig. 2c).

The characteristics of work hardening for samples subjected to tensile deformation at different temperatures are analyzed based on the stress-strain curves using the Hollomon formula (5) [43]:

$$\sigma = K\varepsilon^n \tag{5}$$

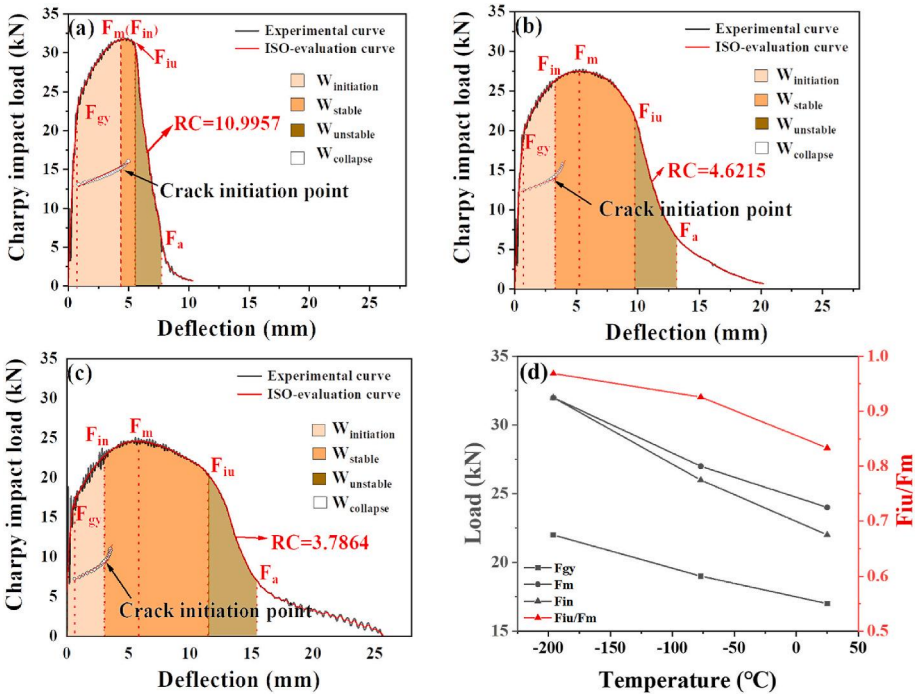


Fig. 9. Typical smoothed impact load-deflection curves: (a) −196 °C; (b) −77 °C; (c) RT; (d) corresponding characteristic values associated with impact properties changing with testing temperature.

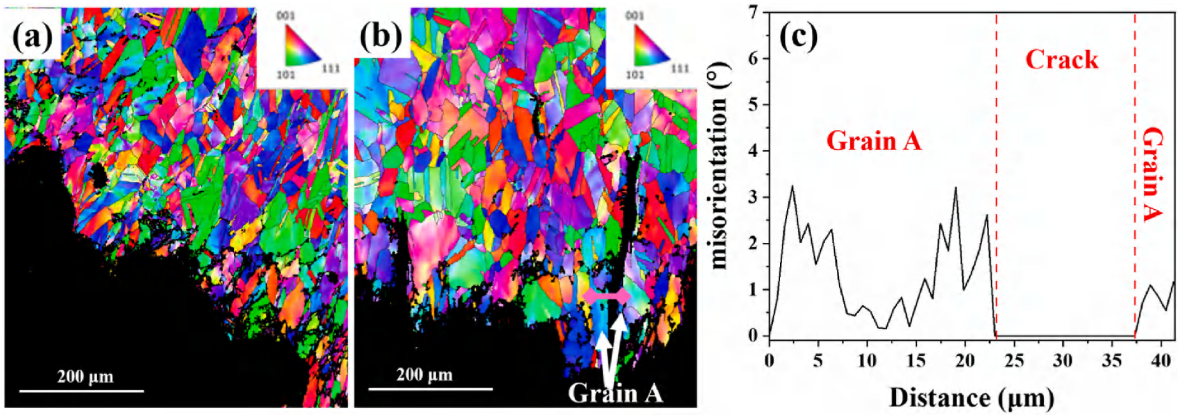


Fig. 10. EBSD analyses for the microstructure near fracture: (a) −196 °C; (b) RT; (c) misorientation analysis for the region near fracture.

where n represents the work hardening exponent and K is the strength coefficient. By taking the natural logarithm of both sides of equation (5), equation (6) can be obtained:

$$ln\,\sigma = lnK + nln\varepsilon \tag{6}$$

Fig. 8 plots the lnσ-lnε curves at different tensile temperatures. It can be observed that the work hardening behavior of the experimental steels can be divided into two stages, where n_1 represents the slope in low strain region and n_2 represents the slope in high strain region. In the earlier stages of uniform plastic deformation corresponding to lower strain, dislocation motion results in dislocation entanglement and pile-up, leading to work hardening [40]. At lower temperatures, the reduced SFE not only contributes to the planarity of the dislocation glide but also hampered cross-slip, resulting in reduced dynamic recovery and increased strain hardening capability. Consequently, the work hardening exponent n_1 increases with decreasing temperature. At higher strain, extensive deformation induces a high density of slip bands, which are commonly referred to as microbands in austenitic low-density steels [27]. These microbands serve as barriers that impede dislocation motion, significantly enhancing the work hardening rate [44]. At lower temperatures, a higher n_2 value indicates a more substantial strengthening effect from microbands. Besides, the presence of mechanical twins also plays an important role in the improved work hardening at −196 °C [45].

4.2. Cryogenic impact toughness

According to the ISO evaluation procedure [46], the experimental curves obtained at −196 °C, −77 °C and RT are processed and separately presented in Fig. 9a-c. The characteristic values associated with impact properties can be obtained based on the ISO evaluation curves. The general yield load (F_{gy}) is at the transition point from elastic deformation to plastic deformation. The maximum Charpy impact load (F_m) corresponds to the peak load of the load-deflection curves.

According to "compliance changing rate (CCR)" method introduced in Ref. [46], CCR can be calculated by equation (8):

$$\frac{\Delta C}{C} = \frac{C - C_{el}}{C_{el}} \tag{8}$$

The compliance C (mm/N) can be calculated using equation (9):

$$C = \frac{ds}{dF} \tag{9}$$

where s and F represent the deflection and load of the load-deflection curve, respectively. The elastic compliance C_{el} (mm/N) is defined as the inverse of the slope of the linear elastic segment of the load-deflection curve, given by equation (10):

$$C_{el} = \frac{s_{el}}{F_{el}} \tag{10}$$

Crack initiation leads to a slope change of CCR curve shown in Fig. 9, where the corresponding load is referred as crack initiation load (F_{in}).

For the load-deflection curve obtained at −196 °C, F_{in} is almost equal to F_m because no obvious change happens to the slope of compliance changing rate curve. The start of sharp near-linear decrease in the load-deflection curve is related to the beginning of unstable crack growth and the corresponding load is referred as F_{iu}. The load corresponding to the end of unstable crack growth is referred as F_a. The remaining section after F_a corresponds to the final collapse stage, unrelated to the impact cracking mechanism [47]. The changes of F_{gy}, F_m, F_{in} and F_{iu}/F_m with testing temperature are plotted in Fig. 9d. It can be found from Fig. 9d that the change of F_{gy} shows a negative dependence of temperature, which is in good agreement with the increase in YS with the decreasing temperature. In addition, as temperature decreases, F_m increases, indicating that the experimental steel could withstand larger impact load at lower temperatures. A larger F_{in} at lower temperature indicates that it is more difficult for crack initiation. The value of F_{iu}/F_m could reflect the development of cracks. A larger F_{iu}/F_m at lower temperature suggests that cracks rapidly propagate from stable to unstable regime. Rebound compliance (RC) is the absolute value of the curve slope from F_{iu} to F_a (as shown in Fig. 9a-c), which reflects the unstable crack growth rate. It can be seen that RC is larger at lower temperature, which is a hint that unstable cracks propagate more rapidly at lower temperature.

The absorbed impact energy associated with different stages of crack initiation and growth, i.e., crack initiation ($W_{initiation}$), stable crack growth (W_{stable}), unstable crack growth ($W_{unstable}$), and final collapse ($W_{collapse}$), are analyzed following the methods in Refs [46], as shown in Fig. 9. The $W_{initiation}$ at −196 °C, −77 °C and RT is 112, 68 and 57 J, respectively. The increasing $W_{initiation}$ with decreasing temperature reveals that the resistance of the experimental steel to crack initiation is stronger at lower temperature. A high level of $W_{initiation}$ is the key factor for maintaining good cryogenic impact toughness [11]. Additionally, as temperature decrease, obvious decreases in both W_{stable} and $W_{unstable}$ can be observed, implying that the resistance to crack propagation is weaker at lower temperature. According to the above, the experimental steel exhibits a higher crack initiation energy upon cryogenic impact, which contributes to good cryogenic toughness. However, the lower crack propagation energy reduces the overall cryogenic toughness of the experimental steel.

The microstructure of the cross-section near fracture surface was analyzed by EBSD, as presented in Fig. 10. For samples tested at −196 °C and RT, the primary crack propagation paths at the fracture surface exhibit a zigzag shape. Most regions exhibit the characteristic of transgranular fracture. Misorientation analysis along the marked line for grain A is presented in Fig. 10c. The misorientation difference between

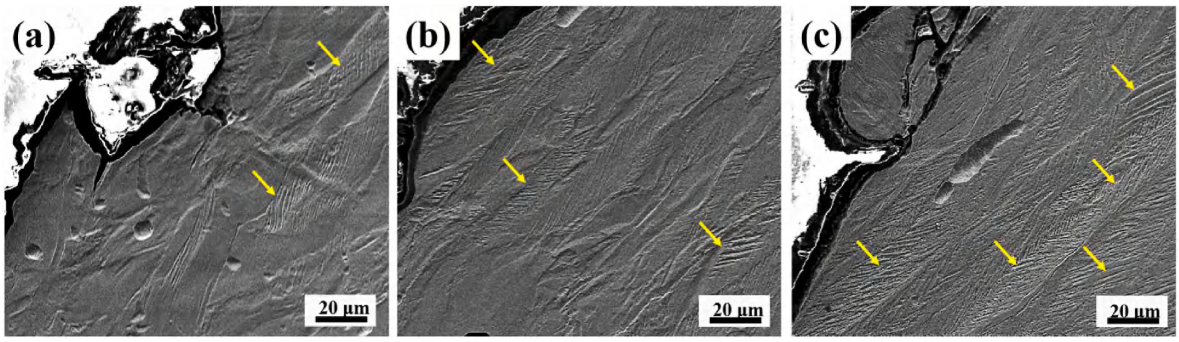


Fig. 11. SEM micrograph for the regions near fracture after impact tests at (a) −196 °C; (b) −77 °C; (c) RT.

the regions on both sides of the crack is less than 15°, implying that the two regions are in the same grain before the crack occurred. This suggests that transgranular fracture occurred in this region. In this case, crack propagation leads to relatively severe plastic deformation near the primary cracks. As a result, a significant local orientation gradient occurs in the region adjacent to the primary cracks within the grain interiors. Besides, secondary cracks present in the sample tested at RT, which is different from that at −196 °C. The presence of secondary cracks release stress concentration at the tip of the primary cracks, making stress redistribute within material. Hence, impact fracture is delayed, contributing to a better toughness [47,48]. This explains that the crack propagation energy is higher at RT.

It is worth noting that the experimental steel exhibits a higher tensile elongation but a lower toughness as deformation temperature decreases. The SFE reduces with the decreasing temperature, increasing glide planarity during low strain rate deformation. A high glide planarity benefits continuous uniform deformation, therefore increasing the tensile elongation [36]. On the other hand, the experimental steel exhibited a poor cryogenic impact toughness, which is opposite to the finding by Liu et al. [49] that FCC-structured high-entropy alloys show higher toughness at cryogenic temperature. The mechanical twinning occurs easily in high-entropy alloys because of lower SFE. The poor cryogenic impact toughness of the experimental steel is related to the higher activation energy for slip systems at lower temperature [29]. Although a lower temperature is corresponding to a higher glide planarity at a low strain rate, activating planar glide is difficult at low temperature during high strain rate deformation because of the higher active energy [29]. As shown in Fig. 11, the characteristic of planar glide can be observed in the regions near fracture. Noted that only few regions experienced planar glide during impact deformation at −196 °C, as shown in Fig. 11a. As temperature increases, planar glide becomes more pronounced. These planar glide features could retard the propagation of crack, thus increasing toughness [50,51]. Therefore, the resistance to crack propagation is weaker at lower temperature.

5. Conclusions

The temperature dependence of the deformation behavior and mechanical properties of the Fe–20Mn–6Al-0.6C-0.15Si austenitic low-density steel was studied through tensile test and Charpy impact test at −196 °C, −77 °C and RT, followed by microstructure analyses. Based on the experimental results, the following conclusions can be drawn.

(1) As the deformation temperature decreases, the SFE decreases, enhancing the planar glide of dislocations. As a result, the LAGB fraction increases with the decreasing temperature. The work hardening rate is higher at lower temperatures. Therefore, both the tensile strength and elongation increase with the decreasing temperature.
(2) As temperature decreasing, the larger F_{in} and higher $W_{initiation}$ indicating a higher resistance to crack initiation. However, a larger F_{iu}/F_m suggests the rapid transition from stable to unstable crack propagation. Consequently, the crack propagation energy is lower at lower temperature, demonstrating a lower resistance to crack propagation. Because the rapid unstable crack propagation, the W_{all} is the lowest at −196 °C.
(3) The smaller fracture surface area, larger fibrous zone and more obvious ductile fracture characteristics in radial zone suggest that the toughness is better at higher temperature. Secondary cracks are observed near the fracture after impact test at RT, which benefits the toughness.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

[1] Bai S, Chen Y, Liu X, Lu H, Bai P, Li D, et al. Research status and development prospect of Fe–Mn–C–Al system low-density steels. J Mater Res Technol 2023;25:1537–59. https://doi.org/10.1016/j.jmrt.2023.06.037.
[2] Chen S, Rana R, Haldar A, Ray RK. Current state of Fe-Mn-Al-C low density steels. Prog Mater Sci 2017;89:345–91. https://doi.org/10.1016/j.pmatsci.2017.05.002.
[3] Hu S, Zheng Z, Yang W, Yang H. Fe–Mn–C–Al low-density steel for structural materials: a review of alloying, heat treatment, microstructure, and mechanical properties. Steel Res Int 2022:2200191. https://doi.org/10.1002/srin.202200191.
[4] Rana R. Low-density steels. JOM 2014;66:1730–3. https://doi.org/10.1007/s11837-014-1137-2.
[5] Chen G, Xu G, Biermann H, Mola J. Microstructure and mechanical properties of Co-added and Al-added austenitic stainless steels. Mater Sci Eng A 2022;854:143832. https://doi.org/10.1016/j.msea.2022.143832.
[6] Curtze S, Kuokkala V-T. Dependence of tensile deformation behavior of TWIP steels on stacking fault energy, temperature and strain rate. Acta Mater 2010;58:5129–41. https://doi.org/10.1016/j.actamat.2010.05.049.
[7] Wang HH, Meng L, Luo Q, Sun C, Li GQ, Wan XL. Superior cryogenic toughness of high-Mn austenitic steel by welding thermal cycles: the role of grain boundary evolution. Mater Sci Eng A 2020;788:139573. https://doi.org/10.1016/j.msea.2020.139573.
[8] Zhang J, Li G, Wang H, Wan X, Hu M, Cao Y. Achieving superior cryogenic impact toughness and sufficient tensile properties in a novel high-Mn austenitic steel weld metal via cerium addition. J Mater Res Technol 2023;23:5016–30. https://doi.org/10.1016/j.jmrt.2023.02.108.
[9] Tang L. Revealing deformation mechanisms of advanced face-centred cubic alloys at cryogenic environment via in situ neutron diffraction. University of Birmingham; 2022.

[10] Chen J, Ren J, Liu Z. Deformation microstructures as well as strengthening and toughening mechanisms of low-density high Mn steels for cryogenic applications. J Mater Res Technol 2021;13:947–61. https://doi.org/10.1016/j.jmrt.2021.05.018.
[11] Wang Y, Wang H, Su Y, Xu P, Shinohara T. Cryogenic impact fracture behavior of a high-Mn austenitic steel using electron backscatter diffraction and neutron Bragg-edge transmission imaging. Mater Sci Eng A 2023;887:145768. https://doi.org/10.1016/j.msea.2023.145768.
[12] Luo Q, Wang HH, Li GQ, Sun C, Li DH, Wan XL. On mechanical properties of novel high-Mn cryogenic steel in terms of SFE and microstructural evolution. Mater Sci Eng A 2019;753:91–8. https://doi.org/10.1016/j.msea.2019.02.093.
[13] Ke R, Wan X, Zhang Y, Hu C, Wu K. The impact of annealing temperature on the microstructure - properties relationship of reversion-induced austenitic stainless steels. Mater Sci Eng A 2022;843:143100. https://doi.org/10.1016/j.msea.2022.143100.
[14] Scherbring S, Chen G, Veltel B, Bartzsch G, Richter J, Vollmer M, et al. Microstructural constituents and mechanical properties of low-density Fe-Cr-Ni-Mn-Al-C stainless steels. Materials 2022;15:5121–40. https://doi.org/10.3390/ma15155121.
[15] De Cooman BC, Estrin Y, Kim SK. Twinning-induced plasticity (TWIP) steels. Acta Mater 2018;142:283–362. https://doi.org/10.1016/j.actamat.2017.06.046.
[16] Chen G, Rahimi R, Xu G, Biermann H, Mola J. Impact of Al addition on deformation behavior of Fe–Cr–Ni–Mn–C austenitic stainless steel. Mater Sci Eng A 2020;797:140084. https://doi.org/10.1016/j.msea.2020.140084.
[17] Kuhlmann-Wilsdorf D. Chapter 59 the LES theory of solid plasticity. In: Nabarro FRN, Duesbery MS, editors. Dislocations solids, vol. 11. Elsevier; 2002. p. 211–342. https://doi.org/10.1016/S1572-4859(02)80010-9.
[18] Otani LB, Vidilli AL, Zepon G, Coury FG, Botta WJ, Uhlenwinkel V, et al. The effect of C on the mechanical behavior of a low-density high-Mn steel. J Mater Res 2024;39:77–89. https://doi.org/10.1557/s43578-023-01162-y.
[19] Kim H, Suh D-W, Kim NJ. Fe–Al–Mn–C lightweight structural alloys: a review on the microstructures and mechanical properties. Sci Technol Adv Mater 2013;14:014205. https://doi.org/10.1088/1468-6996/14/1/014205.
[20] Yoo JD, Park K-T. Microband-induced plasticity in a high Mn–Al–C light steel. Mater Sci Eng A 2008;496:417–24. https://doi.org/10.1016/j.msea.2008.05.042.
[21] Wei LL, Gao GH, Kim J, Misra RDK, Yang CG, Jin XJ. Ultrahigh strength-high ductility 1 GPa low density austenitic steel with ordered precipitation strengthening phase and dynamic slip band refnement. Mater Sci Eng A 2022;838:142829.
[22] Welsch E, Ponge D, Hafez Haghighat SM, Sandlöbes S, Choi P, Herbig M, et al. Strain hardening by dynamic slip band refinement in a high-Mn lightweight steel. Acta Mater 2016;116:188–99. https://doi.org/10.1016/j.actamat.2016.06.037.
[23] Wang H, Cao Z, Gao Z, Wang C, Liang J, Godfrey A, et al. Synergetic strengthening from dynamic slip band-grain boundary interaction in a low-density FeMnAlC steel. Mater Sci Eng A 2023;862:144498. https://doi.org/10.1016/j.msea.2022.144498.
[24] Yoo JD, Hwang SW, Park K-T. Origin of extended tensile ductility of a Fe-28Mn-10Al-1C steel. Metall Mater Trans A 2009;40:1520–3. https://doi.org/10.1007/s11661-009-9862-9.
[25] Luo KT, Gan D. Low temperature mechanical properties of Fe-28Mn-5Al-1C alloy n. d.
[26] Zhang Q, Chen G, Xue Z, Chen Z, Xu G. Correlation between grain size, mechanical properties and deformed microstructure of Fe-20Mn-6Al-0.6C-0.15Si low-density steel. J Iron Steel Res Int 2024. https://doi.org/10.1007/s42243-024-01276-5.
[27] Gutierrez-Urrutia I, Shibata A, Tsuzaki K. Microstructural study of microbands in a Fe-30Mn-6.5Al-0.3C low-density steel deformed at cryogenic temperature by combined electron channeling contrast imaging and electron backscatter diffraction. Acta Mater 2022;233:117980. https://doi.org/10.1016/j.actamat.2022.117980.
[28] Gutierrez-Urrutia I, Shibata A. Effect of deformation temperature on strain localization phenomena in an austenitic Fe-30Mn-6.5Al-0.3C low-density steel. Acta Mater 2024;264:119566. https://doi.org/10.1016/j.actamat.2023.119566.
[29] Hirth JP, Lothe J. Theory of dislocations. second ed.). vol. 50. Krieger Publishing; 1992.
[30] Li W, Liaw PK, Gao Y. Fracture resistance of high entropy alloys: a review. Intermetallics 2018;99:69–83. https://doi.org/10.1016/j.intermet.2018.05.013.
[31] Curtze S, Kuokkala V-T, Oikari A, Talonen J, Hänninen H. Thermodynamic modeling of the stacking fault energy of austenitic steels. Acta Mater 2011;59:1068–76. https://doi.org/10.1016/j.actamat.2010.10.037.
[32] Mola J. Considerations in the design of formable austenitic stainless steels based on deformation-induced processes. IntechOpen; 2017.
[33] Park K-T, Jin KG, Han SH, Hwang SW, Choi K, Lee CS. Stacking fault energy and plastic deformation of fully austenitic high manganese steels: effect of Al addition. Mater Sci Eng A 2010;527:3651–61. https://doi.org/10.1016/j.msea.2010.02.058.
[34] Martin S, Wolf S, Martin U, Krüger L, Rafaja D. Deformation mechanisms in austenitic TRIP/TWIP steel as a function of temperature. Metall Mater Trans A 2016;47:49–58. https://doi.org/10.1007/s11661-014-2684-4.
[35] Huang Q, Volkova O, Biermann H, Mola J. Tensile elongation of lean-alloy austenitic stainless steels: transformation-induced plasticity versus planar glide. Mater Sci Technol 2017;33:1224–30. https://doi.org/10.1080/02670836.2016.1277091.
[36] Rahimi R, Ullrich C, Klemm V, Rafaja D, De Cooman BC, Biermann H, et al. Influence of Al on the temperature dependence of strain hardening behavior and glide planarity in Fe–Cr–Ni–Mn–C austenitic stainless steels. Mater Sci Eng A 2016;649:301–12. https://doi.org/10.1016/j.msea.2015.10.005.
[37] Guo P, Qian L, Meng J, Liu S, Zhang F. Effect of Al on the low cycle fatigue deformation behavior and microstructure evolution of Fe–22Mn-0.6C TWIP steel. J Mater Res Technol 2024;30:2086–98. https://doi.org/10.1016/j.jmrt.2024.03.221.
[38] Zhi H, Zhang C, Antonov S, Yu H, Guo T, Su Y. Investigations of dislocation-type evolution and strain hardening during mechanical twinning in Fe-22Mn-0.6C twinning-induced plasticity steel. Acta Mater 2020;195:371–82. https://doi.org/10.1016/j.actamat.2020.05.062.
[39] Dai Y, Tang D, Mi Z, Lü J. Microstructure characteristics of an Fe-Mn-C TWIP steel after deformation. J Iron Steel Res Int 2010;17:53–9. https://doi.org/10.1016/S1006-706X(10)60142-2.
[40] Wei LL, Gao GH, Kim J, Misra RDK, Yang CG, Jin XJ. Ultrahigh strength-high ductility 1 GPa low density austenitic steel with ordered precipitation strengthening phase and dynamic slip band refinement. Mater Sci Eng A 2022;838:142829. https://doi.org/10.1016/j.msea.2022.142829.
[41] Ma T, Gao J, Li H, Li C, Zhang H, Li Y. Microband-induced plasticity in a Nb content Fe–28Mn–10Al–C low density steel. Metals 2021;11:345. https://doi.org/10.3390/met11020345.
[42] Kumar D. Design of high manganese steels: calculation of SFE and ms temperature. HSLA steels 2015 microalloying 2015 offshore eng. Steels 2015. Cham: Springer International Publishing; 2016. p. 857–63. https://doi.org/10.1007/978-3-319-48767-0_107.
[43] Babich VK, Pirogov VA, Vakulenko IA. Strain-hardening parameters determined from the stress-strain curve. Strength Mater 1977;9:704–7. https://doi.org/10.1007/BF01537770.
[44] Ma Y, Zhang Y, Sun L, Zhang Z. Refractory high-entropy alloys with ultra-high strength and strain hardening ability induced by heterogeneous deformation and microbands. J Mater Sci Technol 2024;192:215–27. https://doi.org/10.1016/j.jmst.2023.12.057.
[45] Tang L, Wang L, Wang M, Liu H, Kabra S, Chiu Y, et al. Synergistic deformation pathways in a TWIP steel at cryogenic temperatures: in situ neutron diffraction. Acta Mater 2020;200:943–58. https://doi.org/10.1016/j.actamat.2020.09.075.
[46] Alexopoulos ND, Stylianos A, Campbell J. Dynamic fracture toughness of Al–7Si–Mg (A357) aluminum alloy. Mech Mater 2013;58:55–68. https://doi.org/10.1016/j.mechmat.2012.11.005.
[47] Alexopoulos ND, Stylianos A. Impact mechanical behaviour of Al–7Si–Mg (A357) cast aluminum alloy. The effect of artificial aging. Mater Sci Eng A 2011;528:6303–12. https://doi.org/10.1016/j.msea.2011.04.086.
[48] Tong L. Low temperature impact toughness of high strength structural steel. 2018.
[49] Liu D, Yu Q, Kabra S, Jiang M, Forna-Kreutzer P, Zhang R, et al. Exceptional fracture toughness of CrCoNi-based medium- and high-entropy alloys at 20 kelvin. Science 2022;378:978–83. https://doi.org/10.1126/science.abp8070.
[50] Yang DC, Jo YH, Ikeda Y, Körmann F, Sohn SS. Effects of cryogenic temperature on tensile and impact properties in a medium-entropy VCoNi alloy. J Mater Sci Technol 2021;90:159–67. https://doi.org/10.1016/j.jmst.2021.02.034.
[51] Cheng J, Mu Y, Zu G, Yao G. Impact toughness and fractography in Mg-Y alloy. Mater Des 2017;123:64–8. https://doi.org/10.1016/j.matdes.2017.03.034.